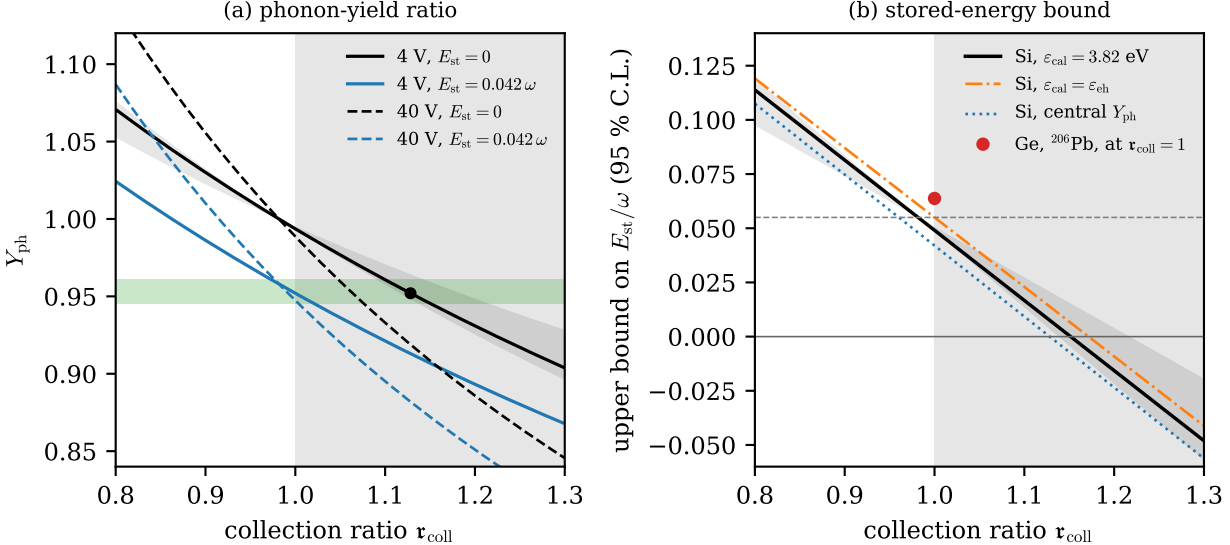
(a) phonon-yield ratio
4 V, $E_{\rm st} = 0$
4 V, $E_{\rm st} = 0.042\,\omega$
40 V, $E_{\rm st} = 0$
40 V, $E_{\rm st} = 0.042\,\omega$
$Y_{\rm ph}$
1.10
1.05
1.00
0.95
0.90
0.85
0.8
0.9
1.0
1.1
1.2
1.3
collection ratio $\mathfrak{r}_{\rm coll}$
(b) stored-energy bound
Si, $\varepsilon_{\rm cal} = 3.82$ eV
Si, $\varepsilon_{\rm cal} = \varepsilon_{\rm eh}$
Si, central $Y_{\rm ph}$
Ge, ^{206}Pb, at $\mathfrak{r}_{\rm coll} = 1$
upper bound on $E_{\rm st}/\omega$ (95 % C.L.)
0.125
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
0.8
0.9
1.0
1.1
1.2
1.3
collection ratio $\mathfrak{r}_{\rm coll}$